About Simulation Evidence & Reliance Boundary Standard - (SERBS)

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Architecture: SIMULOS® — Simulation Governance Architecture Category: Governance & Enforcement Subcategory: Simulation Governance Parent Standard: Simulation Evidence & Reliance Boundary Standard (SERBS) Governed Space: Simulation Evidence & Reliance Boundary

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Canonical Definition

Simulation Evidence & Reliance Boundary Standard (SERBS) governs what simulation-derived information may legitimately become evidence, what evidential meaning and weight that evidence may carry, how far reliance upon it may extend, which uses must remain limited, and how simulation evidence must interface with broader validation governance.

SERBS addresses a fundamental simulation problem:

A simulation can produce technically precise, repeatable, and convincing results without those results necessarily providing equally strong evidence about reality.

The Standard creates the governance layer between simulation results and reliance upon simulation results.

Why SERBS Matters

Organizations increasingly use simulation to support decisions that cannot be tested easily, safely, economically, or repeatedly in the real world.

This creates a methodological risk.

A simulation may contain millions of executions, high-fidelity models, sophisticated physics, realistic environments, and highly precise outputs while still depending upon assumptions, abstractions, scenario limitations, incomplete reality representation, uncertain correlations, or conditions that do not fully transfer to the real world.

SERBS prevents simulation sophistication from being mistaken for evidential certainty.

It requires organizations to establish:

What is evidence? What does that evidence actually support? How much evidential weight may it carry? Under which conditions does that meaning remain valid? Where does legitimate reliance stop? What additional validation is required before broader conclusions are made?

Governance Flow

Governed Simulation Outputs + Simulation-to-Reality Correlation → Simulation Evidence Qualification → Simulation Evidence Weight → Simulation Reliance Boundary → Simulation Use Limitation → Simulation Validation Interface → Governed Simulation Evidence & Reliance Record

Where broader validation is required:

Governed Simulation Evidence & Reliance Record → VALIDOS® Validation Governance

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What SERBS Helps Organizations Govern

SERBS provides a structured method for determining whether simulation-derived information can legitimately support a particular engineering, scientific, safety, mission, operational, or deployment proposition.

It helps organizations distinguish between:

Simulation Output ≠ Simulation Evidence

Simulation Evidence ≠ Validation Determination

Evidence Volume ≠ Evidential Strength

Correlation ≠ Reality Equivalence

Repeated Runs ≠ Independent Evidence

Technical Precision ≠ Evidential Certainty

The result is not less reliance on simulation.

It is better-governed reliance on simulation.

Reliance Boundary

SERBS requires simulation evidence to remain bounded by the conditions that created its meaning.

Those boundaries may include: Simulation Object and configuration; simulation purpose and scope; represented reality; assumptions; model applicability; input basis; scenario coverage; execution conditions; output classification; simulation-to-reality correlation; uncertainty; temporal relevance; known limitations.

Evidence may therefore be highly useful while still being legitimately narrow.

SERBS prevents narrow evidence from silently becoming a broad claim.

Boundary with VALIDOS®

SERBS does not determine whether an entire system, object, claim, or operational proposition has been validated.

The architectural boundary is explicit:

SIMULOS® determines what simulation-derived evidence can legitimately mean.

VALIDOS® determines whether that evidence, alone or together with other evidence, is sufficient to support a Validation Determination or intended reliance.

This allows simulation evidence to enter broader validation without losing the conditions, uncertainty, provenance, and limitations that define its legitimate meaning. Use Case 1 — Space Mission Simulation: Evidence Before Reality Exists

A space organization is preparing a future lunar landing mission.

Large simulation programs reproduce spacecraft dynamics, navigation, landing trajectories, crew interaction, surface operations, failure conditions, environmental effects, and mission contingencies.

Thousands or millions of simulation executions may demonstrate apparently stable mission performance.

But a fundamental problem remains:

the operational reality being predicted may not yet have been experienced in the exact mission configuration being simulated.

Some environmental effects can only be approximated. Real mission data may be sparse. Crew behavior under actual lunar conditions may not yet exist. A highly sophisticated simulator may therefore provide powerful evidence without providing complete real-world proof.

Without SERBS, an organization could gradually transform:

“The mission architecture performed successfully across these governed simulations.”

into:

“The mission architecture is proven to perform successfully in lunar operations.”

Those statements are not methodologically equivalent.

SERBS requires the simulation evidence to carry its actual basis:

tested mission configuration → represented lunar conditions → assumptions → scenario coverage → executions → observed outputs → available reality correlation → remaining uncertainty → permitted reliance.

The resulting evidence may legitimately support mission design, contingency planning, crew preparation, engineering decisions, or a broader validation program while still carrying an explicit boundary against claims that the simulation alone cannot establish.

The value of SERBS is not to weaken the simulation evidence. It is to prevent that evidence from being asked to prove more than the simulation actually tested. Use Case 2 — Autonomous Vehicle or Robot: Simulation Success Before

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Real-World Transfer

An organization develops an autonomous vehicle, mobile robot, delivery robot, industrial robot, or other autonomous system primarily through large-scale simulation.

The system completes millions of simulated scenarios involving traffic, pedestrians, obstacles, sensor conditions, failures, environmental variation, unusual events, and behavioral edge cases.

The simulation campaign reports extremely high success rates.

The organization now considers the system ready for broader real-world deployment.

But another fundamental problem appears:

success inside a simulated environment does not automatically establish equivalent behavior after transfer into physical reality.

Real sensors may respond differently. Friction and dynamics may vary. Lighting, weather, surfaces, object appearance, human behavior, sensor noise, actuator behavior, latency, hardware faults, and combinations of rare conditions may differ from the simulated representation.

Even millions of successful executions can share the same underlying model assumptions or representation gaps.

SERBS therefore asks a different question from:

“How many simulated scenarios did the system pass?”

It asks:

“What real-world proposition may those passed simulations legitimately support?”

SERBS qualifies the evidence, evaluates its simulation-specific weight, identifies common dependencies across the simulation campaign, establishes the permitted reliance boundary, and preserves use limitations before the results enter broader validation or deployment decisions.

The organization may conclude that the evidence strongly supports:

performance across the represented scenario space

while not yet supporting:

equivalent performance across unrestricted operational reality.

Additional real-world testing, observation, correlation, or other validation evidence can then be added through VALIDOS®.

A million successful simulations can increase confidence. They cannot eliminate an ungoverned reality gap.

Enterprise Value

SERBS gives organizations a repeatable governance structure for turning simulation results into decision-usable evidence without overstating what that evidence proves.

It is particularly valuable where: real-world testing is expensive; failure is dangerous; physical testing cannot cover enough scenarios; the operational environment does not yet exist; simulations operate at enormous scale; simulation is used to support safety or deployment decisions; simulation outputs feed regulatory or validation processes; sim-to-real transfer remains uncertain.

SERBS therefore provides the missing boundary between what simulation can show and what an organization may legitimately conclude from what simulation has shown.

Foundational Principle

Simulation-derived evidence must not travel farther than its governed meaning can legitimately support.

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Canonical Source

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